Md. Abdullah Al Mamun

Graduate Researcher, ECE, Temple University • Quantum Algorithms, Quantum ML & Quantum Error Correction

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RESEARCH PROFILE

Quantum computing researcher focused on **when quantum models actually help and what limits them**: trainability and noise diagnostics for variational circuits on **IBM Quantum hardware**, GPU-scale studies of **trainability versus classical simulability** with CUDA-Q, and **noise attribution for early-fault-tolerant neutral-atom protocols**. Author of **HilbertBench**; contributor to QuEra's open-source **tsim** sampler; lead author in deep learning for computational pathology; former industry AI engineer.

EDUCATION

Temple University, Philadelphia, PA Jan 2025 – Present

M.S. in Electrical & Computer Engineering *(expected Dec 2026; one course remaining)* — **GPA 3.78 / 4.00**

Ph.D. program in Electrical Engineering (M.S. en route)

University of Rajshahi, Bangladesh — B.Sc. in Computer Science & Engineering 2018 – 2023

Thesis: Seizure detection from EEG (ResNet; 50+ controlled experiments)

PUBLICATIONS & MANUSCRIPTS

1. **M. A. Al Mamun**, et al., "HilbertBench: A Backend-Agnostic Framework for Quantum Machine Learning Diagnostics with Validation on Quantum Hardware," manuscript in preparation for *Quantum*, 2026. IN PREP
2. S. Purba, **M. A. Al Mamun**, I. Obeid, & J. Picone, "No Quantum Advantage Without Structure: A Controlled Benchmark of Quantum Machine Learning on Classical Data," *IEEE Signal Proc. in Medicine & Biology Symp. (SPMB)*, Dec. 2026. ACCEPTED
3. **M. A. Al Mamun**, S. Purba, I. Obeid, & J. Picone, "Parameter-Efficient Fine-Tuning of Pathology Foundation Models for Microsegmentation of Breast Whole-Slide Images in Digital Pathology," *IEEE SPMB*, Dec. 2026. ACCEPTED
4. **M. A. Al Mamun**, R. C. Purba, I. Obeid, & J. Picone, "Automated Interpretation and Visualization in Digital Pathology," *Proc. 52nd Northeast Bioengineering Conference (NEBEC)*, 2026.
5. B. Hackel, A. Bagritsevich, A. Dumitrescu, **M. A. Al Mamun**, R. C. Purba, D. Heathcote, I. Obeid, & J. Picone, "Enabling Microsegmentation: Digital Pathology Corpora for Advanced Model Development," in *Signal Processing in Medicine and Biology*, Springer, 2026.
6. C. Dumitrescu, S. Purba, **M. A. Al Mamun**, et al., "Rapid and Inexpensive Precision Breast Cancer Screening Using Machine Learning," 2025.

QUANTUM RESEARCH & OPEN-SOURCE SOFTWARE

Star on Atoms — which noise channel limits transversal STAR on neutral atoms Oct 2026 – Present

github.com/mamuncseru/star-on-atoms • pre-registered protocols • tsim, Bloqade, Stim

- Attributed the injection error of **QuEra's transversal STAR architecture** (post-selected small-angle rotations for early fault tolerance) to individual physical noise channels, using QuEra's published circuits and hardware model.
- Committed an exact first-order fault enumeration as a **pre-registered prediction**; confirmed it blind on **802 million sampled shots** at code distances 3, 5 and 7, with 95% Wilson intervals.
- Single-qubit gates carry 43–52% of the injection error; halving their error buys about $1.3\times$ in simulation volume at $d = 5$.
- Found a decoding pitfall and contributed the fix upstream: **QuEraComputing/tsim pull request #234**, merged Nov 2026.

HilbertBench — non-intrusive diagnostics for quantum ML experiments 2025 – Present

github.com/mamuncseru/hilbertbench • hilbertbench.readthedocs.io • DOI 10.5281/zenodo.21009244 • `pip install hilbertbench`

- Designed and built a framework that wraps a single Qiskit V2 **Estimator/Sampler** or PennyLane device and records every circuit execution into an append-only, SHA-256-sealed trace, with no changes to algorithm code and no extra shots.
- Implemented six analyzers computed from the trace: **barren-plateau** detection (gradient variance with CIs), shot-noise ratio, optimizer convergence, circuit structure, **KL expressibility**, and hardware noise profile.
- Ran diagnostics on **IBM Quantum processors** via `EstimatorV2(mode=backend)`; 319 documented tests, CI, four end-to-end tutorials on real QML failure modes; basis of a journal manuscript for *Quantum*.
- Built a **CUDA-Q integration** and ran a GPU pilot on an **NVIDIA A40**: 19,200 recorded executions (4–26 qubits, fp64). Linear-depth local-cost circuits lost gradient variance only about $4\times$ from 4 to 26 qubits while the MPS bond dimension needed to reproduce them rose from 8 to 64 (8 to 20 qubits); recording overhead under 0.5%.

Controlled Benchmark of QML vs. Classical Learning (co-author, IEEE SPMB 2026) 2026

- Benchmarked quantum machine learning models against strong classical baselines on classical data under matched, controlled

conditions, showing that performance gains require problem structure the quantum model can exploit.

Quantum Advantage Quest — self-directed study of quantum-advantage claims 2026 – Present

github.com/mamuncseru/quantum-advantage-quest • project site

- Problem-first search for quantum advantage with pre-registered kill criteria; every claim is attacked with the strongest classical tools first (e.g., dequantization, BBBV lower bounds).
- 13 algorithm "autopsies" (Deutsch–Jozsa → Decoded Quantum Interferometry, quantum Gibbs sampling) and a Problem-Shape Catalog mapping problem structures to quantum mechanisms and to what kills them.
- Tested statevector simulator (NumPy/SciPy): Shor factoring 35 on 19 qubits, glued-trees quantum walk separation, Davies-generator Gibbs sampler, QSVT Chebyshev identity.

RESEARCH EXPERIENCE

Graduate Research Assistant — Temple University Jun – Aug 2025; Jan 2026 – Present

- Quantum ML for the NSF FET project *Quantum Computing Based Approach to Undirected Generative ML Models*: QML benchmarking, trainability diagnostics, and hardware experiments (Qiskit, PennyLane).
- Lead author on parameter-efficient fine-tuning of pathology foundation models for microsegmentation of gigapixel breast whole-slide images (IEEE SPMB 2026); applying reinforcement learning to digital pathology.
- Built GPU-accelerated WSI pipelines (EfficientNet, TransMIL, self-supervised learning) on SLURM HPC clusters.

Research Intern — Kyushu Institute of Technology, Japan Feb 2023 – Mar 2023

Sakura Science Exchange Program. EEG acquisition under controlled protocols; preprocessing and analysis in MATLAB.

INDUSTRY EXPERIENCE

AI Engineer — My MedicalHub International Ltd, Dhaka, Bangladesh Aug 2023 – Dec 2024

- Led real-time computer-vision models for medical diagnostics (classification, detection, segmentation, pose estimation) in PyTorch/TensorFlow; deployed and scaled on AWS and Azure with monitoring and retraining workflows.

Research Assistant (part-time, remote) — My MedicalHub International Ltd, Tampa, FL Aug 2020 – Jul 2023

Built and annotated medical datasets; object detection, segmentation, and pose estimation with systematic evaluation.

TEACHING & COMMUNITY

Instructor — IEEE Quantum Machine Learning Workshop (QMLW 2026) Summer 2026

Designed and delivered hands-on QML lectures and coding exercises (Qiskit, PennyLane) for signal-processing and biomedical applications.

Graduate Teaching Assistant — Temple University Jan 2025 – Dec 2025

Engineering Computation I & II (ECE 1111, ECE 3822); Electrical Devices & Systems I (ECE 2112).

Attendee — IEEE Quantum Week 2026 (QCE26), Toronto, Canada Sep 2026

Participant — Microsoft Quantum QDK Workshop, New York, NY May 2026

TECHNICAL SKILLS & HONORS

Quantum: Qiskit (V2 primitives, IBM Quantum hardware), PennyLane, CUDA-Q (`nvidia`, `tensornet-mps`), Stim, tsim, Bloqade, Microsoft QDK; variational circuits, barren plateaus, expressibility, shot-noise and hardware-noise analysis, QEC simulation and detector error models, quantum walks, Gibbs sampling, QSVT, dequantization.

Machine learning: PyTorch, PyTorch Lightning, TensorFlow; foundation models & parameter-efficient fine-tuning, multiple-instance learning, self-supervised learning, computer vision, reinforcement learning.

Scientific computing & tools: Python, NumPy, SciPy, SymPy, pandas, C++, MATLAB; pytest, Git/GitHub Actions, Docker, Linux, SLURM/HPC, CUDA, MkDocs/Read the Docs, Quarto, LaTeX, AWS, Azure.

AI-native workflow: coding agents (e.g., Claude Code), checked by pre-registered predictions, tests, and logged agent use.

Honors: Sakura Science Exchange Program, JST (2023) • Microdegree in AI, Fusemachines (2022–23) • IEEE Member